Annotated References

1. Low Volatility and Low Beta

Ang, A., Hodrick, R. J., Xing, Y., & Zhang, X. (2006). The cross-section of volatility and expected returns. *The journal of finance*, *61*(1), 259-299.

> *The direct basis for our ivol60 factor. Because the premium comes mainly from the short leg, the excess return achievable by a long-only portfolio is expected to be smaller than the long-short spread.*

Frazzini, A., & Pedersen, L. H. (2014). Betting against beta. *Journal of financial economics*, *111*(1), 1-25.

> *The theoretical basis for our beta60 factor. Its beta estimator, which estimates correlations and volatilities separately and shrinks the estimate toward 1, can be used to reduce the noise in 60-day betas.*

Fama, E. F., & French, K. R. (1992). The cross-section of expected stock returns. *the Journal of Finance*, *47*(2), 427-465.

> *After controlling for size, market beta shows almost no relation to average returns, while size and book-to-market explain the cross-sectional variation in returns. This provides the premise for a low-beta strategy and supports evaluating strategies by alpha after controlling for size and value.*

2. Momentum and Reversal

Jegadeesh, N., & Titman, S. (1993). Returns to buying winners and selling losers: Implications for stock market efficiency. *The Journal of finance*, *48*(1), 65-91.

> *Buying winners and selling losers over the past 3–12 months earns about 1% per month; the returns dissipate after 12 months and partially reverse. This supports our use of a 12-1 momentum window, skipping the most recent month between the formation and holding periods.*

Lehmann, B. N. (1990). Fads, martingales, and market efficiency. *The Quarterly Journal of Economics*, *105*(1), 1-28.

> *Weekly returns show significant reversals: a losers-minus-winners portfolio earns positive returns in about 90% of weeks, but with very high turnover. This explains why momentum skips the most recent period and suggests that short-term reversal signals should be evaluated net of transaction costs.*

Asness, C. S., Moskowitz, T. J., & Pedersen, L. H. (2013). Value and momentum everywhere. *The journal of finance*, *68*(3), 929-985.

> *Value and momentum premia exist across eight markets and asset classes, and the two are negatively correlated, so combining them markedly raises the Sharpe ratio. This*

> *supports our combination of weak factor signals and suggests an equal-weighted combination of factor groups as a benchmark for Lasso.*

Asness, C. S., Frazzini, A., Israel, R., & Moskowitz, T. J. (2014). Fact, fiction and momentum investing. *Journal of Portfolio Management, Fall*.

> *Argues that the momentum premium is also present on the long side and among large stocks and survives transaction costs, while acknowledging its crash risk. This supports keeping long-only momentum in the Singapore market, where shorting is impractical, and helps explain drawdowns during market rebounds.*

3. Liquidity

Amihud, Y. (2002). Illiquidity and stock returns: cross-section and time-series effects. *Journal of financial markets*, *5*(1), 31-56.

> *Proposes ILLIQ and finds that it is positively related to expected returns both cross-sectionally and over time, with the strongest effect among small stocks. It requires only daily data, which suits the Singapore market; we can also use it as a liquidity filter for the stock universe.*

4. Volatility Management and Factor Combination Methods

Moreira, A., & Muir, T. (2017). Volatility-managed portfolios. *The Journal of Finance*, *72*(4), 1611-1644.

> *Scaling positions by the inverse of last month's realized variance raises the Sharpe ratios of the market and many factors, with the largest effect on momentum. This is the basis for our GARCH volatility targeting; the realized-variance version should serve as a benchmark, compared at the same average volatility.*

Drobetz, W., & Otto, T. (2021). Empirical asset pricing via machine learning: evidence from the European stock market. *Journal of Asset Management*, *22*(7), 507-538.

> *Applies the methods of Gu, Kelly & Xiu (2020) to the European market and finds that nonlinear models outperform OLS, with price trends and valuation ratios as the most important predictors. This supports our use of Lasso regression in a small sample and suggests that Elastic Net could mitigate unstable selection among correlated factors.*

Gu, S., Kelly, B., & Xiu, D. (2020). Empirical asset pricing via machine learning. *The review of financial studies*, *33*(5), 2223-2273.

> *Provides a comparative analysis of machine learning methods for empirical asset pricing and the measurement of asset risk premia. The results show that machine learning forecasts can deliver large economic gains to investors.*

Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of econometrics*, *31*(3), 307-327.

> *Generalizes the ARCH model of Engle (1982) to GARCH, capturing volatility clustering with few parameters. We use a GARCH model to forecast portfolio volatility as the basis for position scaling in volatility targeting.*

Tibshirani, R. (1996). Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, *58*(1), 267-288.

> *Proposes the Lasso, which adds an L1 penalty on the sum of absolute coefficients to least squares so that some coefficients shrink exactly to zero. We use Lasso to select and combine signals from the candidate factors, reducing overfitting in a small sample.*